

Time response

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Mechatronics 3DX4

Slides 5: Time Response

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Introduction

- ▶ Once we have obtained a mathematical representation of a system, our next step is to analyze its transient and steady-state response.
- ▶ In this section, we will focus on how to analyze a system's transient response.
- ▶ We already know how to determine the output response by solving differential equations, or taking inverse Laplace

transforms.

- ▶ These methods are laborious and time consuming.
- ▶ We want to develop a technique where we can get the desired information about a system's transient and steady-state response, basically by inspection.
- ▶ Our first topic will be how to analyze poles and zeros to determine a system's response.

Poles and Zeros

- ▶ Consider:

$$G(s) = \frac{b_{n-1}s^{n-1} + b_{n-2}s^{n-2} + \cdots + b_0}{s^n + a_{n-1}s^{n-1} + \cdots + a_0} = \frac{N(s)}{D(s)}$$

- ▶ **Poles** of $G(s)$ are the roots of $D(s)$.
- ▶ **Zeros** of $G(s)$ are the roots of $N(s)$. $\frac{s+1}{(s+1)(s+2)} \rightarrow s=-2$ pole
- ▶ Generally, at poles $G(s) = \infty$ unless the pole is cancelled by a matching zero.
- ▶ At zeros, $G(s) = 0$ unless the zero is cancelled by a matching pole.

Poles and Zeros of First Order System eg.

- ▶ A system's output response contains two parts:
 1. **Forced** or **steady state response**: this is caused by the poles of the input function $R(s)$

the input function, $x(s)$.

2. **Natural** or **homogeneous response**: this is caused by the poles of the transfer function, $G(s)$.

- In example below, our transfer function is $G(s) = \frac{s+2}{s+5}$, and our input is $R(s) = \frac{1}{s}$.

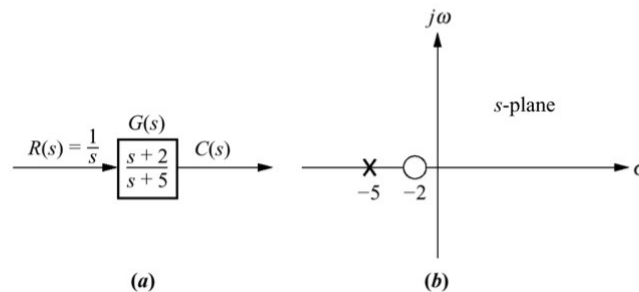


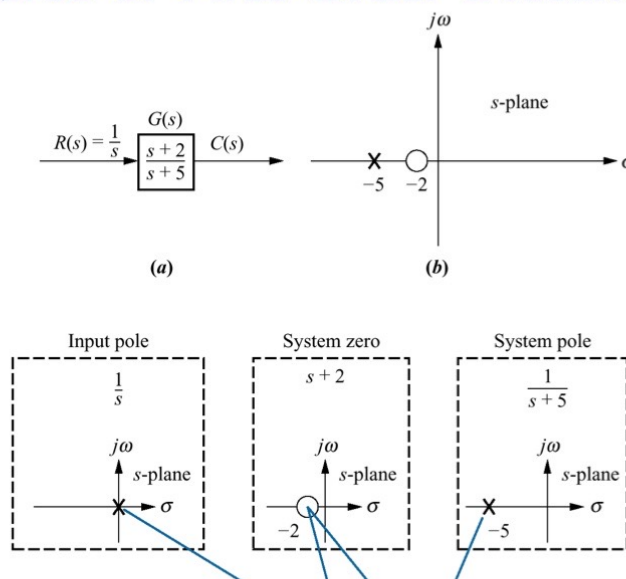
Figure 4.1.

$$\frac{s+2}{s(s+5)} = \frac{A}{s} + \frac{B}{s+5}$$

$$c(t) = (A + Be^{5t})u(t)$$

Forced response 就是 steady state response, 是input function的pole产生的; natural response是transient response, 是系统的transfer function的pole产生的, pole越靠左, 衰减越快。Zero 不会影响衰减速度, 但是会影响响应的形状和幅度。

Poles and Zeros of First Order System eg. - II



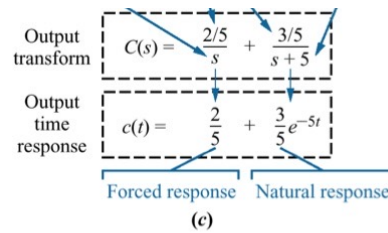


Figure 4.1.

Poles and Zeros of First Order System eg. - III

1. Pole of input function generated forced response, $u(t)$.
2. Pole of transfer function generated natural response, e^{-5t} .

The above is not affected at all by the zero.

3. Pole on real axis, say at $-\alpha$, generates an exponential response, $e^{-\alpha t}$.

Farther to the left on negative axis, the faster the response decays.

4. Both the poles and zeros contribute to the amplitude of the response (ie. the $\frac{2}{5}$ and $\frac{3}{5}$ factors).

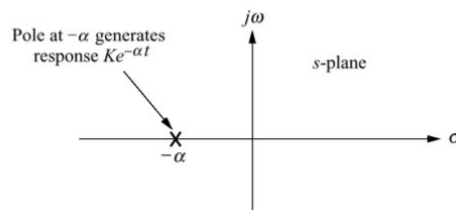


Figure 4.2.

Evaluating Response using Poles eg.

- Consider system shown below.

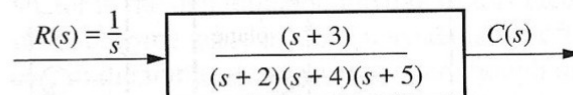


Figure 4.3.

- From inspection, we can immediately determine:

$$C(s) \equiv \underbrace{\frac{K_1}{s}}_{\text{Forced}} + \underbrace{\frac{K_2}{s+2} + \frac{K_3}{s+4} + \frac{K_4}{s+5}}_{\text{Natural}}$$

- Using $\mathcal{L}\left\{\frac{1}{s+\alpha}\right\} = e^{-\alpha t}$ gives:

$$c(t) \equiv \underbrace{K_1}_{\text{Forced}} + \underbrace{K_2 e^{-2t} + K_3 e^{-4t} + K_4 e^{-5t}}_{\text{Natural}}$$

输入的pole会导致forced response, 而系统G的pole会决定natural response。pole在左边的越远, 衰减的越快。
虽然zero不会影响指数衰减项, 但是zero和pole都会影响响应的amplitude。

First Order Systems

- We will now examine first order systems without zeros so we can define performance specifications.
- We use systems of the form $G(s) = \frac{a}{s+a}$ as our base form for our definitions.
- If our input is the step function, $R(s) = \frac{1}{s}$, we get

$$c(t) = c_f(t) + c_n(t) = 1 - e^{-at}$$

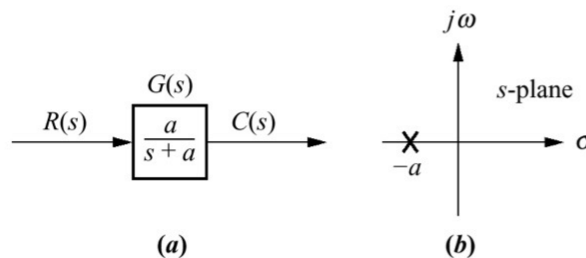


Figure 4.4.

$$G(s) = \frac{k}{s+a} = \frac{k}{a} \frac{a}{s+a}$$

for $a > 0$ pole @ $s = -a$

$$C(s) = \frac{a}{s+a} \frac{1}{s}$$

$$C(\infty) = \lim_{s \rightarrow 0} s C(s) = 1$$

以一个一阶无zero系统为例，响应由forced和natural两部分组成，natural response会逐渐随时间趋于0。

First order system我们把基本模型设置为 $G(s) = \frac{K}{s+a}$ ，如果输入是 step input，那么step response是 $\frac{K}{a}(1 - e^{-at})$

Time Constant

- We will now examine first order systems without zeros so we can define performance specifications.

- ▶ Our first specification is the system's **time constant**, $\tau = \frac{1}{a}$.
- ▶ The time constant is the time required for step response to rise to 63% percent of its final value.

$$c(\tau) = 1 - e^{-a\tau} = 1 - e^{-a \cdot \frac{1}{a}} = 1 - e^{-1} \approx 1 - 0.37 = 0.63$$

- ▶ As $\frac{dc(t)}{dt} = ae^{-at}$, we thus have a equal to the slope at $t = 0$.
- ▶ We call a the **exponential frequency**.

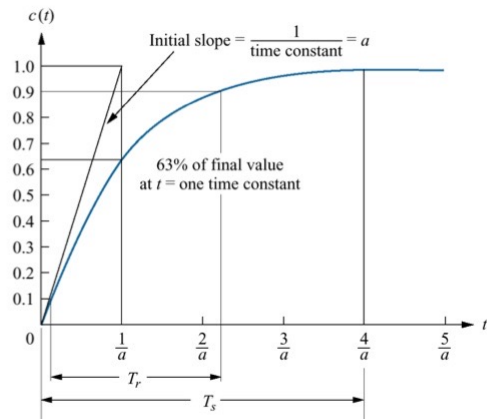


Figure 4.5

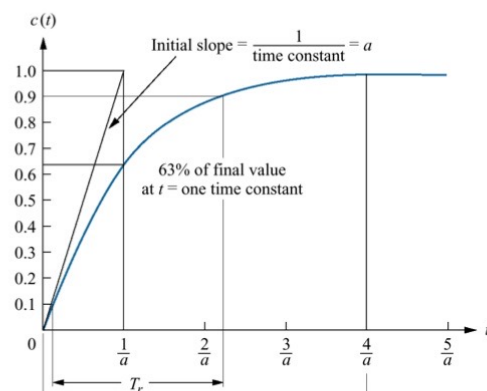
简单来说，time constant tau就是当 $t=\tau$, $y=63\%$ of final steady value.

Rise and Settling Time

- ▶ **Rise time**, T_r , is the time for the output to go from 10% to 90% of its final value.
- ▶ Can show that $T_r = \frac{2.2}{a}$.
- ▶ **Settling time**, T_s , is time required for the output to reach 98% of its final value.

- ▶ Setting $c(T_s) = 0.98$, we find that $T_s = \frac{4}{a}$.

Figure 4.5



Rise time是y从10%上升到90% of final value的时间; settling time是y进入并且一直保持在至少98% of final value的第一次进入时间。

Using Testing to Determine Transfer Function

- ▶ It is quite often not possible or practical to determine a system's transfer function by analytical means.
- ▶ In general, gain of system at $s = 0$ (D.C. gain) is not unity.
- ▶ A more general model would be $G(s) = \frac{K}{s + a}$.
- ▶ Step response is thus

$$C(s) = \frac{K}{s(s + a)} = \frac{\frac{K}{a}}{s} - \frac{\frac{K}{a}}{s + a}$$

- ▶ We thus have

$$c(t) = \frac{K}{a}(1 - e^{-at})$$

- ▶ How can we experimentally determine the values of K and a ?

我们可以根据响应曲线反推G。我们这里假设用一阶无zero系统，上图是时域响应和对应的频域G。

这里得到的c(t)是在step input下一阶系统的响应解。

Using Testing to Determine Transfer Function - II

- ▶ For system to be a first order system, its unit step response

- ▶ For system to be a first-order system, its unit step response should have *no overshoot* and should have a *nonzero initial slope*, as in diagram below.
- ▶ From diagram, we can see the final value is about 0.72, thus 63% of that is $0.63 \times 0.72 = 0.45$.
- ▶ From diagram, the output reaches 0.45 at about $\tau = 0.13$ (time constant).
- ▶ We thus have $a = \frac{1}{\tau} = 7.7$.
- ▶ We next note that
$$c(\infty) = \frac{K}{a}(1 - e^{-at})|_{t \rightarrow \infty} = \frac{K}{a}$$
- ▶ Thus $K = a \cdot c(\infty) = (7.7)(0.72) = 5.54$
- ▶ Thus $G(s) = \frac{5.54}{s + 7.7}$

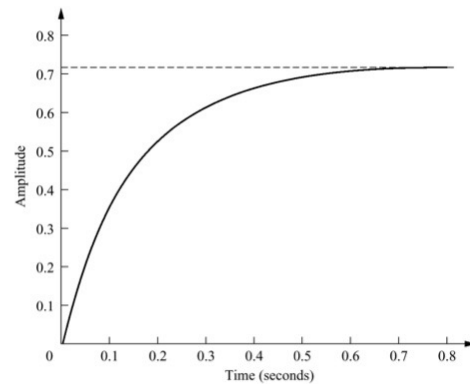
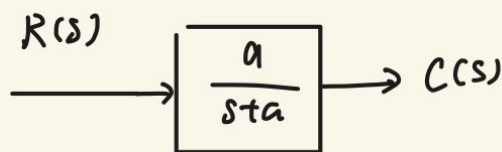


Figure 4.6

一个一阶系统的step response应该没有overshoot，同时初始斜率不是0。

如何通过响应图像写G:

1. 先确定final steady value，乘以63%，找到此时对应的t值，确定tau。
2. 由于 $\tau = 1/a$ ，那么我们可以算出衰减系数a，因为 K/a 是系数，根据final value算出K。



$$G(s) = \frac{a}{s+a}$$

$$\text{step response } C(s) = G(s)R(s)$$

$$= \frac{a}{s(s+a)} = \frac{A}{s} + \frac{B}{s+a}$$

$$\frac{1}{s(s+a)} = \frac{A}{s} + \frac{B}{s+a}$$

cover up method $A = \frac{a}{a} = 1, B = -1$

$$C(s) = \frac{1}{s} - \frac{1}{s+a}$$

$$C(t) = (1 - e^{-at})u(t)$$

$$t_r = t_{0.9} - t_{0.1}$$

In general for $0 < x < 1$

$$C(t_x) = x = 1 - e^{-at_x}$$

$$e^{-at_x} = 1 - x$$

$$-at_x = \ln(1-x)$$

$$t_x = \frac{\ln(1-x)}{-a}$$

$$t_r = t_{0.9} - t_{0.1}$$

$$= \frac{\ln(1-0.9)}{-a} - \frac{\ln(1-0.1)}{-a}$$

$$\approx \frac{2.2}{a}$$

2% of settling time $t_s = t_{0.98} = \frac{\ln(1-0.98)}{-a} = \frac{4}{a}$

这里是推导rise time和settling time的公式。

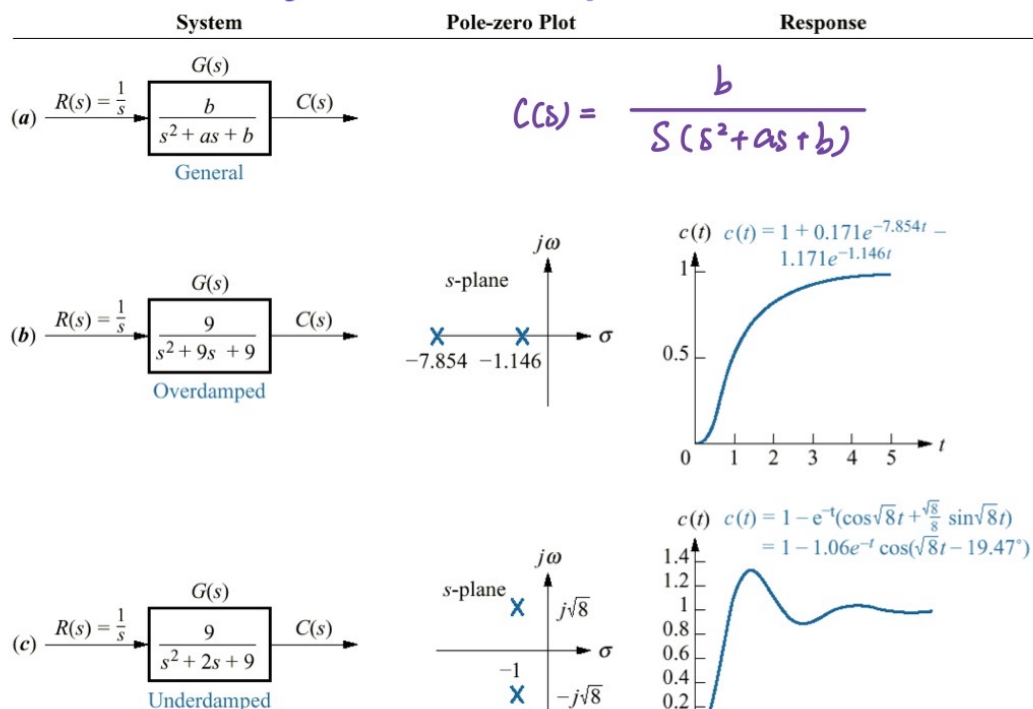
Second-Order Systems

- ▶ For first order systems, varying the systems parameters only changed the speed of the response.
- ▶ Form of a second order system we will analyze is

$$G(s) = \frac{b}{s^2 + as + b} \rightarrow \text{Unity system}$$
- ▶ Changes in these parameters can actually change the *form* of the system's response.
- ▶ May see responses similar to first-order system, damped oscillations, or undamped oscillations.

Unity system指DC gain=1的系统。

Second-Order System Examples



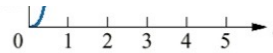


Figure 4.7.

简单来说，overdamped就是两个实数pole，underdamped是两个复数pole。

Second-Order System Examples -II

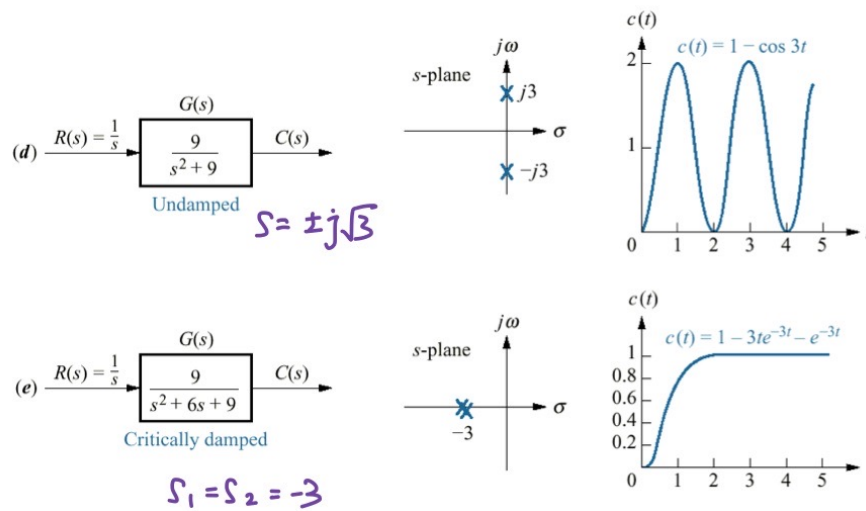


Figure 4.7.

- Like for first-order system, we want to determine information about system's steady state and transient response by examination.

Undamped 是两个纯虚数pole（没有衰减项），critically damped是 overdamped和underdamped分界线，即上升最快的但是没有overshoot，它是两个相同的实数pole。

Overdamped Response

- For **overdamped response**, we have a system with two non equal real poles.

- The unit step response to system below is

$$C(s) = \frac{9}{s(s^2 + 9s + 9)} = \frac{9}{s(s + 7.854)(s + 1.146)}$$

- From inspection of poles, we know form of system's response will be:

$$c(t) = K_1 + K_2 e^{-\sigma_1 t} + K_3 e^{-\sigma_2 t}$$

where $-\sigma_1 = -7.854$ and $-\sigma_2 = -1.146$, are our two real poles.

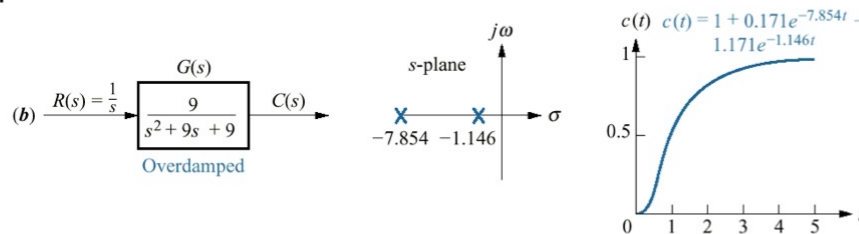


Figure 4.7.

Critically Damped Response

- For **critically damped response**, we have a system with two equal real poles.
- The unit step response to system below is

$$C(s) = \frac{9}{s(s^2 + 6s + 9)} = \frac{9}{s(s + 3)^2} = \frac{K_1}{s} + \frac{K_2}{(s + 3)} + \frac{K_3}{(s + 3)^2}$$

- From inspection of poles, we know form of system's response will be:

$$c(t) = K_1 + K_2 e^{-\sigma_1 t} + K_3 t e^{-\sigma_1 t}$$

where $-\sigma_1 = -3$ is our pole location.

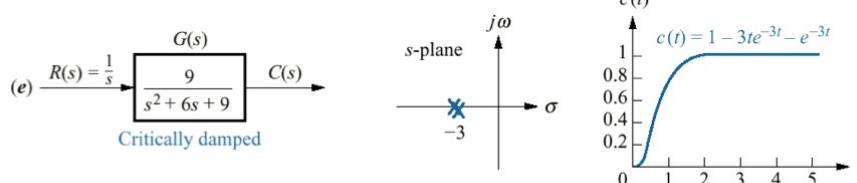


Figure 4.7.

Underdamped Response

- ▶ For **underdamped response**, we have a system with two complex conjugate poles (non zero real and imaginary parts).
- ▶ The unit step response to system below is

$$C(s) = \frac{9}{s(s^2 + 2s + 9)} = \frac{9}{s(s + 1 + j\sqrt{8})(s + 1 - j\sqrt{8})}$$

$$= \frac{K_1}{s} + \frac{\alpha + j\beta}{s + 1 + j\sqrt{8}} + \frac{\alpha - j\beta}{s + 1 - j\sqrt{8}}$$

- ▶ Thus the form of system's response will be:

$$c(t) = K_1 + e^{-\sigma_d t} [2\alpha \cos \omega_d t + 2\beta \sin \omega_d t]$$

where $-\sigma_d \pm j\omega_d = -1 \pm j\sqrt{8}$.

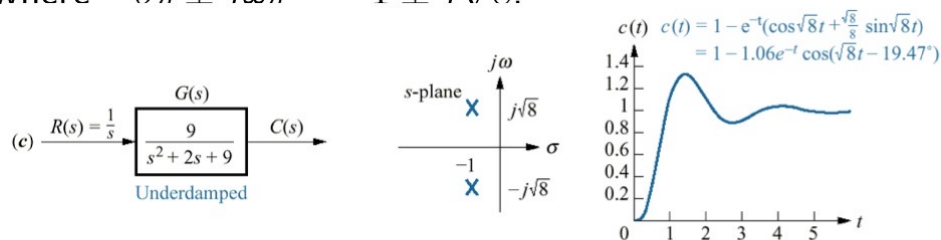
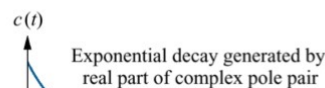


Figure 4.7.

Underdamped Response - II

- ▶ For system with poles at $s = -\sigma_d \pm j\omega_d$, the real part (σ_d) determines the exponential frequency (decay rate) for the exponential envelope.
- ▶ The imaginary part, ω_d , determines the oscillation frequency of the sinusoids, and is called the **damped frequency of oscillation**.
- ▶ Can show that



$$e^{-\sigma_d t} [2\alpha \cos \omega_d t + 2\beta \sin \omega_d t]$$

$$= K_4 e^{-\sigma_d t} \cos(\omega_d t - \phi)$$

where $\phi = \tan^{-1}(\frac{\beta}{\alpha})$ and

$$K_4 = \sqrt{(2\alpha)^2 + (2\beta)^2}$$

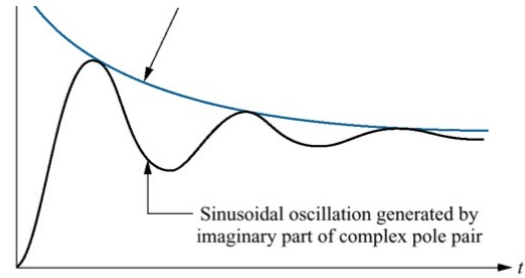


Figure 4.8

Undamped Response

- For **undamped response**, we have a system with two imaginary poles (zero real part).
- The unit step response to system below is

$$C(s) = \frac{9}{s(s^2 + 9)} = \frac{9}{s(s + j3)(s - j3)} = \frac{K_1}{s} + \frac{\alpha + j\beta}{s + j3} + \frac{\alpha - j\beta}{s - j3}$$

- Thus the form of system's response will be:

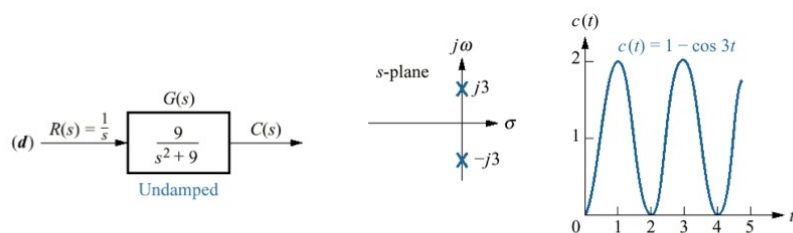
$$c(t) = K_1 + e^{-(0)t} [2\alpha \cos \omega_d t + 2\beta \sin \omega_d t]$$

$$= K_1 + 2\alpha \cos \omega_d t + 2\beta \sin \omega_d t$$

$$= K_1 + K_4 \cos(\omega_d t - \phi)$$

- where
- $$\pm j\omega_d = \pm j3.$$

Figure 4.7



Second-order System's Step Responses

- Critically damped case represents the transition between the underdamped and overdamped cases.
- Critically damped case is the fastest response without

- critically damped case is the fastest response without overshoot.

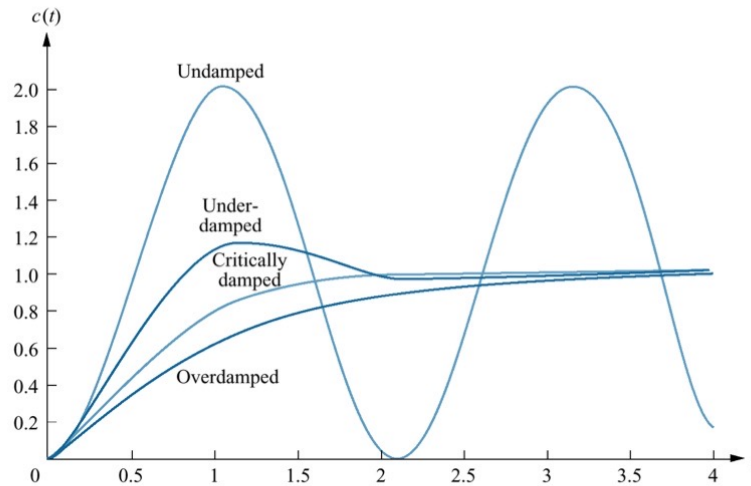


Figure 4.10.

General Second-Order Systems

- We now generalize our discussion of second-order systems and develop specifications to describe the response of the system.
1. The **natural frequency**, ω_n , of a second-order system is the frequency of oscillation of the system with damping removed.
 2. The **damping ratio** of a second-order system is a way to describe a system's damped oscillation, independent of time scale.

We define *damping ratio*, ζ , to be

$$\zeta = \frac{\text{Exponential decay frequency}}{\text{Natural frequency(rad/sec)}}$$

ω_n 就是系统的自然振荡频率, damping ratio是衰减速度/振荡频率,

ζ 系统行为

$\zeta = 0$ 永远振荡

$0 < \zeta < 1$ 振荡但会衰减

$\zeta = 1$ 不振荡, 最快稳定

$\zeta > 1$ 不振荡但变慢

Deriving Parameters - ω_n

- ▶ We want to rewrite the second-order system shown below in terms of ω_n and ζ

$$G(s) = \frac{b}{s^2 + as + b} \quad (1)$$

- ▶ The quadratic equation tells us the poles are:

$$s_{1,2} = \frac{-a \pm \sqrt{a^2 - 4b}}{2} = \frac{-a}{2} \pm \frac{\sqrt{a^2 - 4b}}{2} \quad (2)$$

- ▶ To determine ω_n , we need an undamped system; thus $a = 0$, $G(s) = \frac{b}{s^2 + b}$.
- ▶ Our poles are thus $s_{1,2} = \pm j\sqrt{b}$, giving $\omega_n = \sqrt{b}$ and $b = \omega_n^2$.

Deriving Parameters - ζ

$$G(s) = \frac{b}{s^2 + as + b} \quad s_{1,2} = \frac{-a}{2} \pm \frac{\sqrt{a^2 - 4b}}{2}$$

- For an underdamped system, the poles must have a real part, $\sigma = \frac{-a}{2}$.
- The *exponential decay frequency* is equal to the absolute value of σ .

$$\zeta = \frac{\text{Exponential decay frequency}}{\text{Natural frequency(rad/sec)}} = \frac{|\sigma|}{\omega_n} = \frac{a/2}{\omega_n} \quad (3)$$

- We thus have:

$$a = 2\zeta\omega_n \quad (4)$$

- We can now rewrite our system as

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad (5)$$

Relating Parameters to Poles

- Substituting into the poles equation gives:

$$\begin{aligned} s_{1,2} &= \frac{-a}{2} \pm \frac{\sqrt{a^2 - 4b}}{2} = \frac{-(2\zeta\omega_n)}{2} \pm \frac{\sqrt{(2\zeta\omega_n)^2 - 4\omega_n^2}}{2} \\ &= -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1} \end{aligned} \quad (6)$$

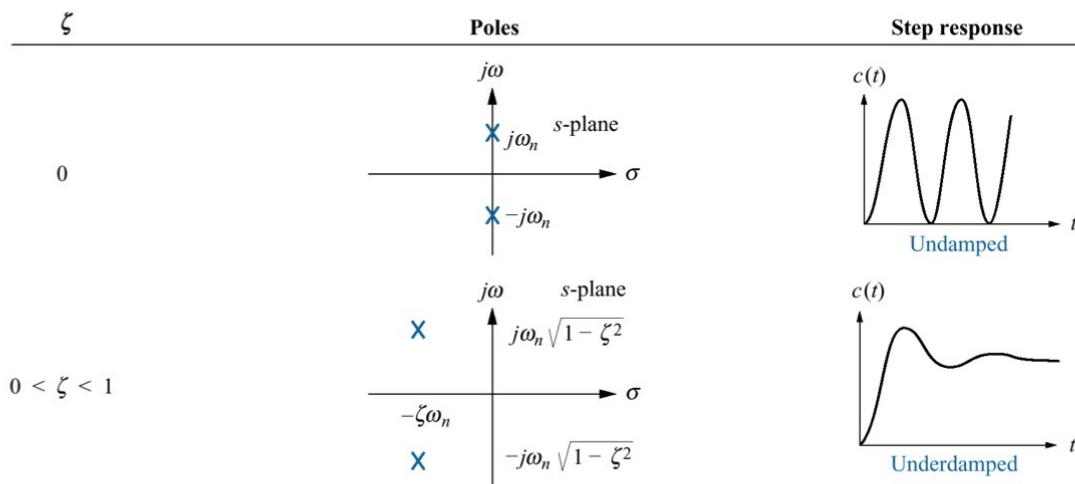


Figure 4.11.

Relating Parameters to Poles - II

$$s_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$$

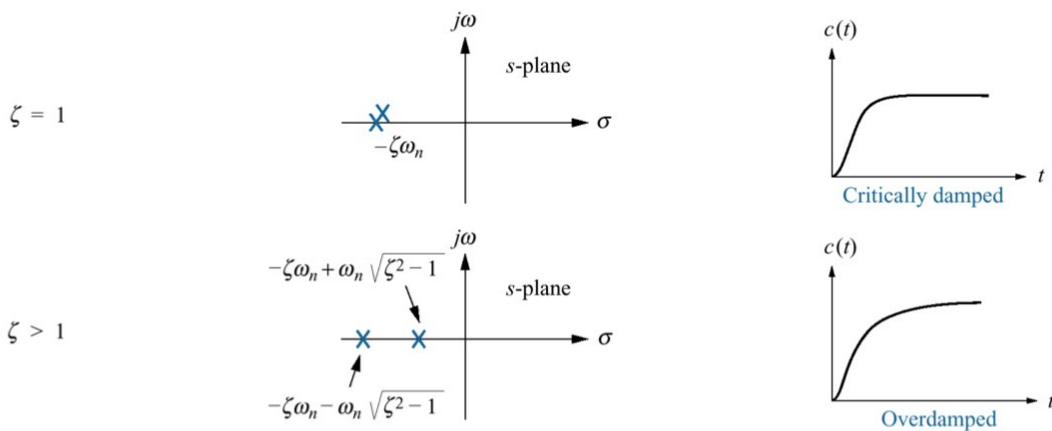


Figure 4.11.

For overdamped, poles $\in$ 2 different real, $\therefore \zeta^2 - 1 > 0$
 $\therefore \zeta > 1$

underdamp, 2 imag poles, $\zeta^2 - 1 < 0$, $0 < \zeta < 1$

critical damp, 2 same real poles, $\zeta^2 - 1 = 0$, $\zeta = 1$

undamp, $\zeta = 0$

Underdamped Second-order Systems

- ▶ We are mostly interested in underdamped systems as they give us the fastest response.
- ▶ Need to examine behavior more closely for analysis and design.
- ▶ Will now define transient specifications for underdamped responses.
- ▶ The step response is:

$$C(s) = \frac{\omega_n^2}{s(s^2 + 2\zeta\omega_n s + \omega_n^2)} = \frac{K_1}{s} + \frac{K_2 s + K_3}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad (7)$$

- ▶ Assuming $\zeta < 1$ (ie. underdamped case), partial fractions gives:

$$C(s) = \frac{1}{s} - \frac{(s + \zeta\omega_n) + \frac{\zeta}{\sqrt{1-\zeta^2}}\omega_n\sqrt{1-\zeta^2}}{(s + \zeta\omega_n)^2 + \omega_n^2(1-\zeta^2)} \quad (8)$$

我们更关注underdamped system, 因为他们有更快的响应。

$$\begin{aligned} C(s) &= \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} - \frac{1}{s} \\ &= \frac{1}{s} - \frac{s + 2\zeta\omega_n}{s^2 + 2\zeta\omega_n s + \omega_n^2} \end{aligned}$$

$$s^2 + 2\zeta \omega_n s + \omega_n^2 = s^2 + 2\zeta \omega_n s + \zeta^2 \omega_n^2 - \zeta^2 \omega_n^2 + \omega_n^2$$

$$= (s + \zeta \omega_n)^2 + \omega_n^2 (1 - \zeta^2)$$

$$C(s) = \frac{1}{s} - \frac{s + \zeta \omega_n + \zeta \omega_n}{(s + \zeta \omega_n)^2 + \omega_n^2 (1 - \zeta^2)}$$

Note: $\mathcal{L}[\cos \omega t] = \frac{s}{s^2 + \omega^2}$, $\mathcal{L}[\sin \omega t] = \frac{\omega}{s^2 + \omega^2}$

$$C(s) = \frac{1}{s} - \left(\frac{s + \zeta \omega_n}{(s + \zeta \omega_n)^2 + \omega_n^2 (1 - \zeta^2)} \right) - \left[\frac{\zeta \omega_n}{(s + \zeta \omega_n)^2 + \omega_n^2 (1 - \zeta^2)} \right]$$

$$\frac{\zeta}{\sqrt{1 - \zeta^2}} \frac{\omega_n \sqrt{1 - \zeta^2}}{(s + \zeta \omega_n)^2 + \omega_n^2 (1 - \zeta^2)}$$

$$c(t) = 1 - e^{-\zeta \omega_n t} (\cos(\omega_n \sqrt{1 - \zeta^2} t)) + \frac{\zeta}{\sqrt{1 - \zeta^2}} \sin(\omega_n \sqrt{1 - \zeta^2} t)$$

Recall trig identity

$$a \cos x + b \sin x = c \cos(x - \varphi)$$

$$c = \sqrt{a^2 + b^2}$$

$$\varphi = \tan^{-1}\left(\frac{b}{a}\right) \quad \text{for } a, b > 0$$

$$a = 1$$

$$b = \frac{\zeta}{\sqrt{1 - \zeta^2}}$$

$$c = \sqrt{1 + \frac{\zeta^2}{1 - \zeta^2}} = \frac{1}{\sqrt{1 - \zeta^2}}$$

$$\varphi = \tan^{-1}\left(\frac{\zeta}{\sqrt{1 - \zeta^2}}\right)$$

$$c(t) = 1 - c e^{-\zeta \omega_n t} \cos(\omega_n \sqrt{1 - \zeta^2} t - \varphi)$$

solving for $i(t)$

Method 2

$$\mathcal{L}\{C'(t)\} = sC(s)$$

$$sC(s) = \frac{1}{s} \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$= \frac{\omega_n^2}{(s + \zeta\omega_n)^2 + \omega_n^2(1 - \zeta^2)}$$

$$= \frac{\omega_n}{\sqrt{1 - \zeta^2}} \left(\frac{\omega_n \sqrt{1 - \zeta^2}}{(s + \zeta\omega_n)^2 + \omega_n^2(1 - \zeta^2)} \right)$$

$$C'(t) = \frac{\omega_n}{\sqrt{1 - \zeta^2}} e^{-\zeta\omega_n t} \sin(\omega_n \sqrt{1 - \zeta^2} t)$$

Underdamped Second-order Systems - II

$$C(s) = \frac{1}{s} - \frac{(s + \zeta\omega_n) + \frac{\zeta}{\sqrt{1 - \zeta^2}}\omega_n\sqrt{1 - \zeta^2}}{(s + \zeta\omega_n)^2 + \omega_n^2(1 - \zeta^2)}$$

- We can now use the inverse Laplace transform below that we derived earlier

$$\mathcal{L}\{K_1 e^{-at} \cos \omega t + K_2 e^{-at} \sin \omega t\} = \frac{K_1(s + a) + K_2 \omega}{(s + a)^2 + \omega^2}$$

- This gives:

$$c(t) = 1 - e^{-\zeta\omega_n t} \left(\cos \omega_n \sqrt{1 - \zeta^2} t + \frac{\zeta}{\sqrt{1 - \zeta^2}} \sin \omega_n \sqrt{1 - \zeta^2} t \right)$$

$$= 1 - \frac{1}{\sqrt{1 - \zeta^2}} e^{-\zeta\omega_n t} \cos(\omega_n \sqrt{1 - \zeta^2} t - \phi) \quad (9)$$

$$\text{where } \phi = \tan^{-1} \left(\frac{\zeta}{\sqrt{1 - \zeta^2}} \right).$$

这是一个zero-free second-order system的通用解， w_d 就是系统在阻尼情况下，实际的振荡频率，其实和理论上undamped的情况下的 w_n 数值上不一样。

Underdamped Second-order Systems - III

$$c(t) = 1 - \frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \cos(\omega_n \sqrt{1-\zeta^2} t - \phi)$$

where $\phi = \tan^{-1}\left(\frac{\zeta}{\sqrt{1-\zeta^2}}\right)$.

- We can now plot the output with the time axis normalized to the natural frequency.

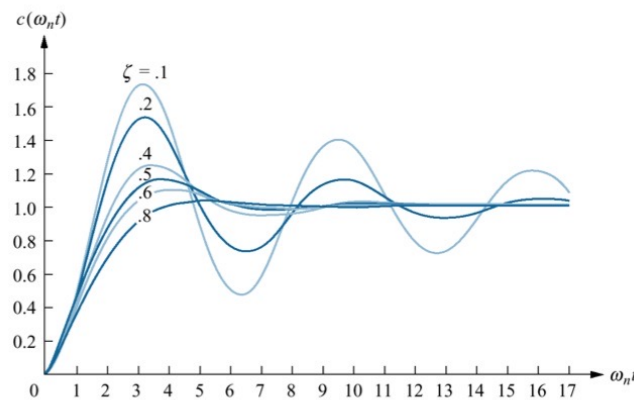


Figure 4.13.

Underdamped Response Specifications

Let $c_{final} = \lim_{t \rightarrow \infty} c(t)$.

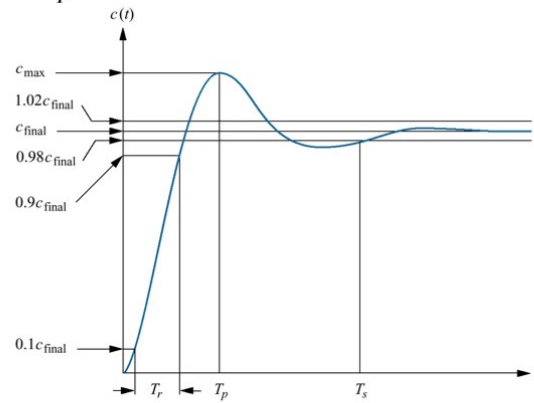
1. **Rise time, T_r** , is the time for the output to go from 10% ($0.1c_{final}$) to 90% ($0.9c_{final}$) of its final value.
2. **Peak time, T_p** , is the time required to reach the first and largest peak, c_{max} .
3. **Percent overshoot, %OS**, is the percentage that the output

overshoots the final value at $t = T_p$.

$$\%OS = \frac{c_{max} - c_{final}}{c_{final}} \times 100\%$$

4. **Settling time, T_s** , is time required for the output to reach and stay within $\pm 2\%$ of c_{final} .

Figure 4.14



注意：之前我们一直用的是unit gain形式，即 $\omega_n^2/(s^2 + 2\zeta\omega_n s + \omega_n^2)$ ，但是如果我们发现就是系统在unit step下的final响应不是1，说明系统乘了constant gain，此时得把 $C(s) = R(s)G(s)$ ，然后做final value theorem看最终值是多少，然后根据图中的信息求解K。

Calculating Peak time

- We can determine T_p by differentiating $c(t)$ in equation 9 and finding the first time it equals zero after $t = 0$.

$$\mathcal{L}\{\dot{c}(t)\} = sC(s) = \frac{\omega_n^2}{(s^2 + 2\zeta\omega_n s + \omega_n^2)} \quad (10)$$

- As we want in form of $\mathcal{L}\{\sin\omega t\}$, we complete the square for the denominator giving

$$\begin{aligned} \mathcal{L}\{\dot{c}(t)\} = sC(s) &= \frac{\omega_n^2}{(s + \zeta\omega_n)^2 + \omega_n^2(1 - \zeta^2)} \\ &= \frac{\frac{\omega_n}{\sqrt{1-\zeta^2}} \omega_n \sqrt{1-\zeta^2}}{(s + \zeta\omega_n)^2 + \omega_n^2(1 - \zeta^2)} \end{aligned} \quad (11)$$

- We thus have

$$\dot{c}(t) = \frac{\omega_n}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin(\omega_n \sqrt{1-\zeta^2} t) \quad (12)$$

Calculating Peak time - II

- ▶ Setting $\dot{c}(t) = 0$ gives

$$\omega_n \sqrt{1 - \zeta^2} t = n\pi$$

thus

$$t = \frac{n\pi}{\omega_n \sqrt{1 - \zeta^2}} \quad (13)$$

- ▶ At $n = 1$, we get first time derivative equals zero after $t = 0$.
We thus have:

$$T_p = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}} \quad (14)$$

注意：T_p就是peak value time，即overshoot最高点的t值，就是第一次响应曲线derivative=0的时候。

Calculating %OS

- ▶ Percent overshoot is: $\%OS = \frac{c_{max} - c_{final}}{c_{final}} \times 100\%$
- ▶ To determine c_{max} , we need to evaluate $c(T_p)$ by substituting equation (14) into equation (9) reproduced below:

$$c(t) = 1 - e^{-\zeta\omega_n t} \left(\cos\omega_n \sqrt{1 - \zeta^2} t + \frac{\zeta}{\sqrt{1 - \zeta^2}} \sin\omega_n \sqrt{1 - \zeta^2} t \right)$$

$$c(T_p) = 1 - e^{-\zeta\pi/\sqrt{1-\zeta^2}} \left(\cos\pi + \frac{\zeta}{\sqrt{1-\zeta^2}} \sin\pi \right)$$

$$c_{\max} = 1 + e^{-\zeta\pi/\sqrt{1-\zeta^2}} \quad (15)$$

- ▶ From equation 9, it is easy to see that $c(\infty) = c_{final} = 1$.
- ▶ Substituting into the %OS formula gives:

$$\%OS = \frac{1 + e^{-\zeta\pi/\sqrt{1-\zeta^2}} - 1}{1} \times 100\%$$

$$\%OS = e^{-\zeta\pi/\sqrt{1-\zeta^2}} \times 100\% \quad (16)$$

Calculating %OS -II

$$\%OS = e^{-\zeta\pi/\sqrt{1-\zeta^2}} \times 100\%$$

- ▶ However, what if we knew which %OS we wanted?
- ▶ We can use the equation above to solve for ζ in terms of %OS. This gives:

$$\zeta = \frac{-\ln(\%OS/100)}{\sqrt{\pi^2 + \ln^2(\%OS/100)}} \quad (17)$$

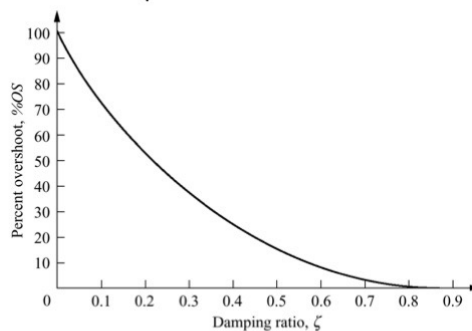


Figure 4.15.

Calculating 2% Settling Time

Calculating Settling Time

- ▶ Settling time is when the output reaches and stays within 2% of its final value.
- ▶ This occurs at latest when the exponential envelope of equation 9 reaches the value of 0.02. This gives:

$$\frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} = 0.02 \quad (18)$$

- ▶ Solving for t gives:

$$\begin{aligned} e^{-\zeta\omega_n t} &= 0.02\sqrt{1-\zeta^2} \\ -\zeta\omega_n t &= \ln(0.02\sqrt{1-\zeta^2}) \\ T_s &= \frac{-\ln(0.02\sqrt{1-\zeta^2})}{\zeta\omega_n} \end{aligned} \quad (19)$$

$$T_s \approx \frac{4}{\zeta\omega_n} \quad (20)$$

注意：settling time的定义是，进入该点之后的区间误差都不能超过2%，所以根据

$$c(t) = 1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}} \sin(\omega_d t + \phi)$$

sin项的振幅必须=0.02。

Calculating Rise Time

- ▶ There does not exist a precise relationship between rise time and damping ratio.
- ▶ Instead, we use a computer and equation 9 to solve for $c(\omega_n t_1) = 0.1c_{final}$ and $c(\omega_n t_2) = 0.9c_{final}$, normalizing for the natural frequency.
- ▶ We then calculate the *normalized rise time*, $\omega_n T_r$, as

$$\omega_n T_r = (\omega_n t_2) - (\omega_n t_1)$$

- ▶ Then, we can use charts below to solve for T_r given a specific ζ and ω_n .

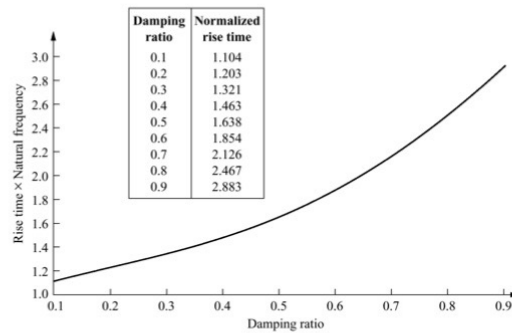


Figure 4.16.

Rise time没有简单公式，于是我们引入normalized time, $\omega_n t$ ，这样可以消除 ω_n 的影响；由于rise time主要受damping ratio的影响，所以我们查表可以得到normalized rise time。

System Response with Additional Poles

- ▶ In last section, we analyzed second-order systems.
- ▶ The formulas we have derived for percent overshoot, settling time, and peak time are only directly valid for systems with two complex poles and no zeros.
- ▶ However, sometimes we can approximate a higher-order system as a second-order system containing the **dominant poles**.
- ▶ The dominant poles are the two poles farthest to the right.

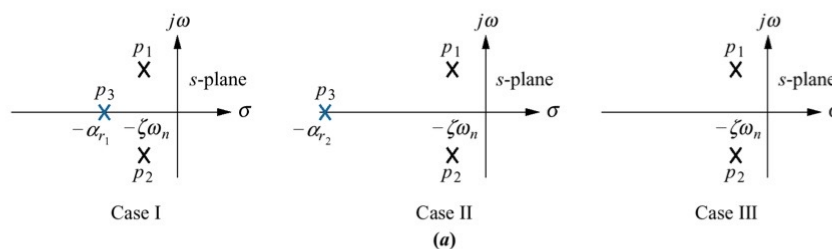


Figure 4.23.

overshoot、settling time、peak time 的公式只适用于两个 complex poles 且没有 zero 的系统，也就是

$$G(s) = \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2}$$

所以在工程上，对于多pole系统，如果有两个pole影响最大，我们可以做近似；在 s plain上，靠右的pole衰减慢，靠左的pole衰减快，所以最靠右的pole决定系统响应。即dominant pole。

System Response with Additional Poles - II

- ▶ How far away do the additional poles have to be?.
- ▶ Depends on the accuracy you want.
- ▶ Text assumes that if a pole is five times more to the left than the dominant poles, then system is represented by the dominant poles.
- ▶ If above met, you would design using the second-order approximation, then simulate final system to make sure it satisfies the design specifications such as %OS, and T_s etc.

如果一个pole比dominant pole至少往左五倍，就可以被忽略。

如果 dominant pole 在：

$\sigma = -\gamma$

s = -2

那么额外极点最好在:

$$s \leq -10$$

因为:

$$5 \times 2 = 10$$

Example Three Pole Systems

► Compare step responses of systems below:

$$T_1(s) = \frac{24.542}{s^2 + 4s + 24.542}$$

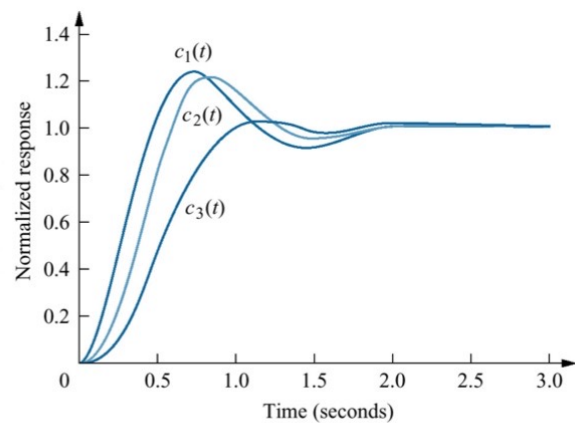
$$T_2(s) = \frac{10 \times 24.542}{(s + 10)(s^2 + 4s + 24.542)}$$

$$T_3(s) = \frac{3 \times 24.542}{(s + 3)(s^2 + 4s + 24.542)}$$

$$c_1(t) = 1 - 1.09e^{-2t} \cos(4.532t - 23.8^\circ)$$

$$c_2(t) = 1 - 0.29e^{-10t} - 1.189e^{-2t} \cos(4.532t - 53.34^\circ)$$

$$c_3(t) = 1 - 1.14e^{-3t} + 0.707e^{-2t} \cos(4.532t + 78.63^\circ)$$



$$1. s = -2 \pm 4.532j$$

$$2. s = -10.$$

$$3. s = -3$$

Justification for Ignoring Nondominant Poles

- ▶ As long as the nondominant pole is far enough to the left, then its contribution to the output will be negligible.
- ▶ Easy to see that this will cause it to decay quickly, but what about its amplitude?
- ▶ Consider the third order system below

$$C(s) = \frac{bc}{s(s^2 + as + b)(s + c)} = \frac{A}{s} + \frac{Bs + C}{s^2 + as + b} + \frac{D}{s + c}$$

- ▶ If we assume steady state response is unity, and that the nondominant pole is at $s = -c$, we can then solve for the following constants using partial fractions:

$$\begin{aligned} A &= 1; & B &= \frac{ca - c^2}{c^2 + b - ca} \\ C &= \frac{ca^2 - c^2a - bc}{c^2 + b - ca}; & D &= \frac{-b}{c^2 + b - ca} \end{aligned} \quad (21)$$

Justification for Ignoring Nondominant Poles - II

$$\begin{aligned} A &= 1; & B &= \frac{ca - c^2}{c^2 + b - ca} \\ C &= \frac{ca^2 - c^2a - bc}{c^2 + b - ca}; & D &= \frac{-b}{c^2 + b - ca} \end{aligned}$$

► If we let $c \rightarrow \infty$, we find:

$$\begin{aligned} A &= 1; & B &= -1 \\ C &= -a; & D &= 0. \end{aligned} \quad (22)$$

这里说明，对于nondominant pole，不仅衰减的快，同时其相应的amplitude也很小，所以可以忽略。

System Response with Zeros

- We now examine systems with Zeros.
- As we saw before, zeros don't change the type of system, but can affect the constants found during partial fraction expansion.
- Consider the system below

$$G(s) = \frac{(s + a)}{(s + b)(s + c)} = \frac{A}{s + b} + \frac{B}{s + c}$$

where by partial fractions we have:

$$A = \frac{-b + a}{-b + c}; \quad B = \frac{-c + a}{-c + b} \quad (23)$$

zero不会改变系统的阶数和类型，但是会改变响应的系数（振幅）。

System Response with Zeros -II

$$A = \frac{-b + a}{-b + c}; \quad B = \frac{-c + a}{-c + b}$$

- ▶ When the zero is far to the left, it will be much larger than the poles, thus:

$$A \approx \frac{a}{-b + c}; \quad B \approx \frac{a}{-c + b}$$

- ▶ Our system then becomes

$$G(s) \approx a \left[\frac{\frac{1}{-b+c}}{s+b} + \frac{\frac{1}{-c+b}}{s+c} \right] = \frac{a}{(s+b)(s+c)}$$

如果系统的zero在s plain上很往左，那么其不会改变系统的特性，即rise time这些，只是改变整体gain。

Nonminimum Phase System

Non-minimum-Phase System

- ▶ What if the zero is in the right half plane (ie. $a < 0$)?
- ▶ If $C(s)$ is the response of a system, then after adding zero at $-a$, we get:

$$(s + a)C(s) = sC(s) + aC(s)$$
- ▶ If the derivative term $sC(s)$ is larger than the scaled response $aC(s)$, the system will initially follow the derivative in the wrong direction!
- ▶ If we take $r(t) = -u(t)$ as our input, we could get a system like the one below.

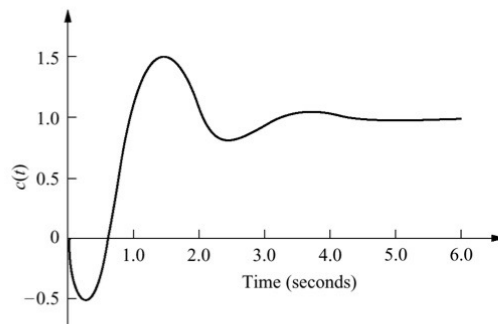


Figure 4.26.

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这页在讲 **Non-minimum phase system (非最小相位系统)**，核心就是：

- 如果系统有 **Right-Half-Plane (RHP) zero**，系统会先往反方向动。

这也是控制里一个非常重要的现象，叫：

Inverse Response (反向响应)

原本系统的输出 $C(s)$ ，那么添加了一个 zero 之后就是 $(s+a)C(s) = sC(s) + aC(s)$ ，对应关系是

s-domain time-domain

$(C(s))$ $(c(t))$

$(sC(s))$ $(c'(t))$

如果 zero 在右边半平面，那么 $a < 0$ ， $c'(t) = |a|c(t)$ ，系统刚开始的时候 $c(t) \approx 0$ ，主要

要由 $C(t)$ 决定，如果input是 $-u(t)$ ，那么初始斜率就是正的，所以一开始会向上震荡。

Pole Zero Cancellations

- Consider the system below:

$$G(s) = \frac{K(s + z_1)}{(s + p_1)(s^2 + as + b)} \quad (24)$$

- If z_1 and p_1 are close enough to each other, they can effectively cancel each other even though they are not exactly equal.

$$C(s) = \frac{26.25(s + 4)}{s(s + 4.01)(s + 5)(s + 6)} \quad (25)$$

$$= \frac{0.87}{s} - \frac{5.3}{s + 5} + \frac{4.4}{s + 6} + \frac{-0.033}{s + 4.01} \quad (26)$$

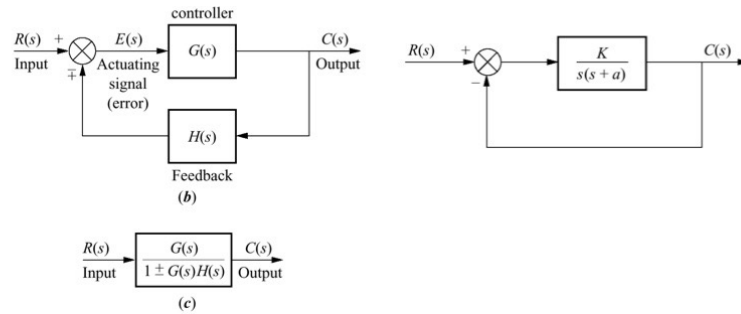
$$\approx \frac{0.87}{s} - \frac{5.3}{s + 5} + \frac{4.4}{s + 6} \quad (27)$$

如果一个pole和zero离得很近，他们会互相cancel。

Analysis and Design of Feedback Systems

- Using the block diagram algebra we developed earlier, we can now apply our second-order system results to feedback systems.
- Applying feedback reduction, we find that the equivalent closed-loop transfer function of system on right is

$$T(s) = \frac{K}{s^2 + as + K} \quad (28)$$



Figures 5.6 and 4.14.

右边的系统经过化简之后就变成了28的样子，注意这是一个标准二阶系统的 transfer function，通过调节K可以改变natural frequency和damping ratio，进而改变rise time, overshoot, settling time这些。

Analysis and Design of Feedback Systems - II

$$T(s) = \frac{K}{s^2 + as + K}$$

- ▶ As we increase K from zero, the poles of the system will go from overdamped ($0 \leq K < \frac{a^2}{4}$), critically damped ($K = \frac{a^2}{4}$), to underdamped ($K > \frac{a^2}{4}$).

$$s_{1,2} = \frac{-a}{2} \pm \frac{\sqrt{a^2 - 4K}}{2} \quad s_{1,2} = \frac{-a}{2} \quad s_{1,2} = \frac{-a}{2} \pm \frac{j\sqrt{4K - a^2}}{2}$$

Gain Design for Transient Response

- Design the value for system gain, K , such that the system response has 10% overshoot.
- Applying feedback reduction, our closed-loop transfer function becomes

$$T(s) = \frac{K}{s^2 + 5s + K} \quad (29)$$

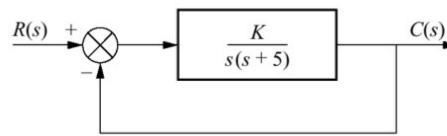
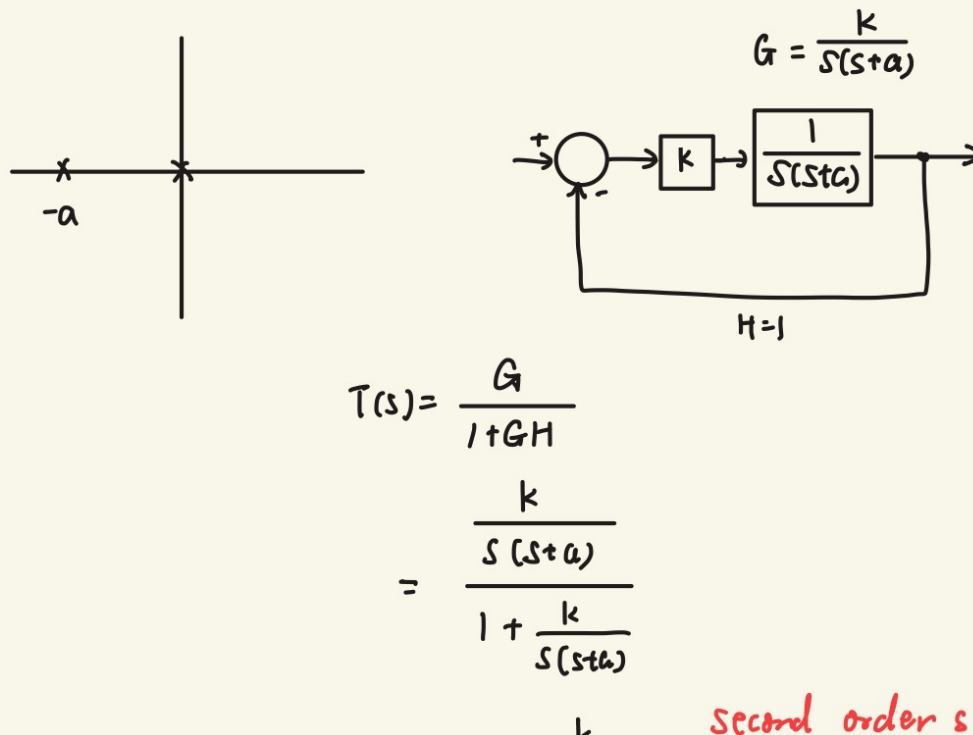


Figure 5.16.



$$T(s) = \frac{K}{s^2 + as + k} \quad \text{with unity gain}$$

Recall: $G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$

$$\omega_n^2 = k$$

$$\omega_n = \sqrt{k}$$

$$2\zeta\omega_n = a \Rightarrow \text{negative of my pole}$$

$$\zeta = \frac{a}{2\omega_n} = \frac{a}{2\sqrt{k}}$$

$$\sqrt{k} = \frac{a}{2\zeta}$$

$$k = \frac{a^2}{4\zeta^2}$$

if given a, ζ , k can be determined

$$\zeta = \frac{-\ln\left(\frac{0.05}{1.05}\right)}{\sqrt{\pi^2 + \left[\ln\left(\frac{0.05}{1.05}\right)\right]^2}} = \frac{-\ln(0.1)}{\sqrt{\pi^2 + \ln(0.1)^2}} \approx 0.5912$$

$a \leq 5$

$$k = \frac{25}{4 \times (0.5912)^2} = 17.8845$$

Matlab solution to the problem

```
1 a=5
2 G=zpk([], [0 -a], 1)
3 pos=10
4 zeta=-log(pos/100)/sqrt(pi^2+log(pos/100)^2)
5 K=a^2/(4*zeta^2)
6 Wn=sqrt(K)
7
8 rlocus(G)
9 sgrid(zeta, Wn)
10 pause
11
12 Gcl=K*G/(1+K*G)
13 step(Gcl)
14 [y, t]=step(Gcl);
15 max(y)
```